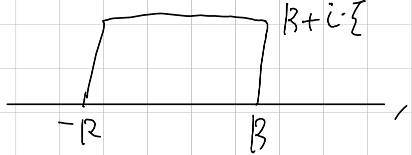
「Evaluate
$$\int_{-\infty}^{\infty} e^{-\pi x^2}\cdot e^{-2\pi i x\xi}\,dx = e^{-\pi\xi^2}. \quad (\xi > 0)$$

Proof. Let $f(z) = e^{-\pi z^2}$. Then, it is entire.

Consider four contour: $C_1 : t \ (-R \le t \le R)$,
$C_2 : R + it \ (0 \le t \le \xi)$
$C_3 : t + i\xi \ (-R \le t \le R)$
$C_4 : -R + it \ (0 \le t \le \xi)$

Then, the composition $C = C_1 \cdot C_2 \cdot (-C_3) \cdot (-C_4)$ is described as



and, by the Cauchy's theorem, $\int_C f = 0$. Now,
$$\int_{C_1} f = \int_{-R}^{R} e^{-\pi x^2}\,dx,$$
$$\int_{C_2} f = \int_0^{\xi} e^{-\pi(R+it)^2}\cdot i\,dt = i\cdot\int_0^{\xi} e^{-\pi(R^2-t^2+2iRt)}\cdot dt$$
and so $\left|\int_{C_2} f\right| = \left|\int_0^{\xi} e^{\pi\cdot(t^2-R^2)}\,dt\right| \le \left|\int_0^{\xi} e^{\pi t^2}\,dt\right| \cdot e^{-\pi R^2} \to 0$ as $R \to \infty$.

Similarly $\left|\int_{C_4} f\right| \to 0$ as $R \to \infty$. And,
$$\int_{C_3} f = \int_{-R}^{R} e^{-\pi(t+i\xi)^2}\,dt = \int_{-R}^{R} e^{-\pi(t^2-\xi^2+2it\xi)}\,dt$$
$$= e^{\pi\xi^2}\cdot\int_{-R}^{R} e^{-\pi t^2}\cdot e^{-2\pi i t\xi}\,dt$$

Thus, as $R \to \infty$,
$$0 = \int_{-\infty}^{\infty} e^{-\pi x^2}\,dx - e^{\pi\xi^2}\int_{-\infty}^{\infty} e^{-\pi t^2}\cdot e^{-2\pi i t\xi}\,dt. \quad \blacksquare$$